

What the score hides

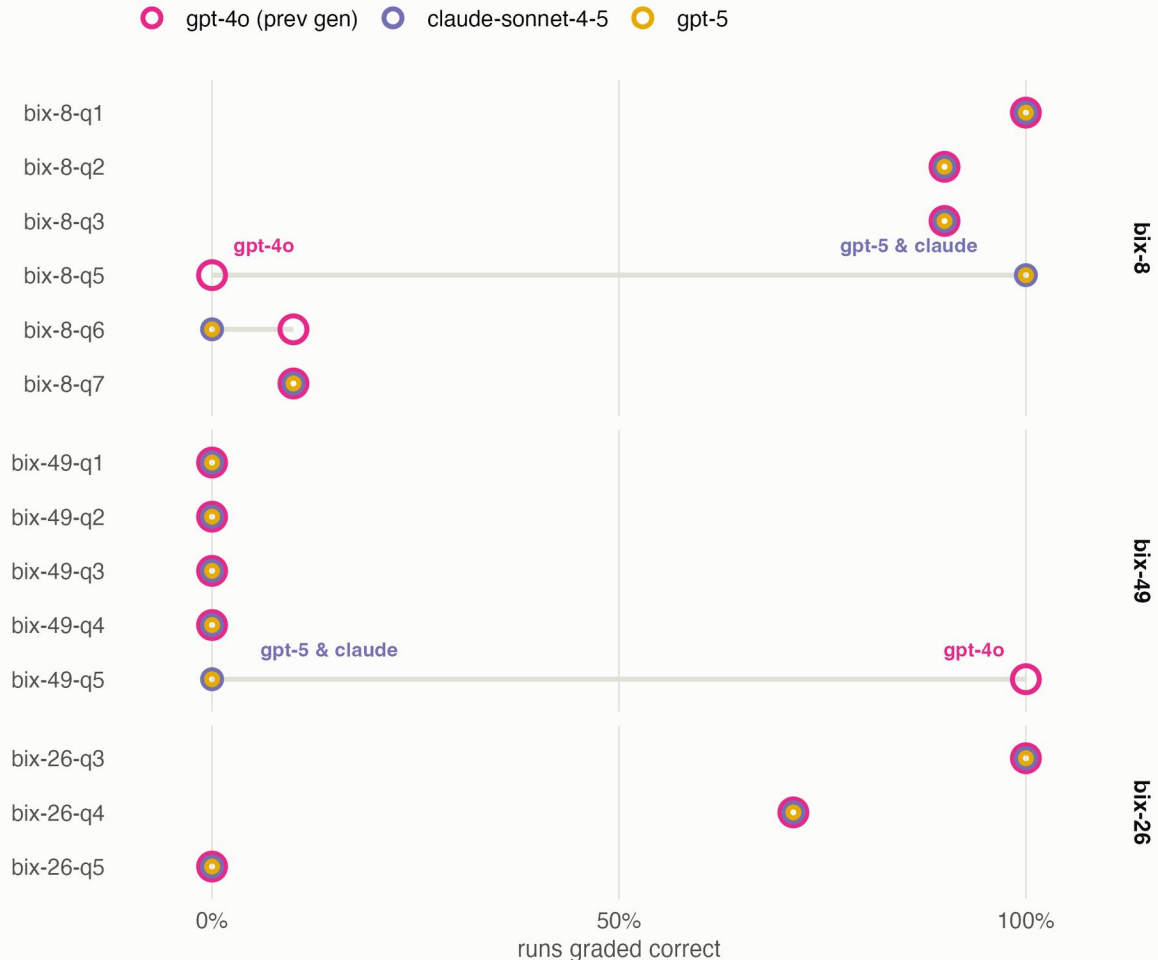
A bioinformatics-agent benchmark scores each question with a single pooled number. I replicated the runs, replicated the grading, and re-derived every answer key to see what that number leaves out.

140	agent runs, 10 per question
3,900	grading verdicts, 3 graders × 10 rounds
14	answer keys re-derived from raw data

FIGURE 1

Grader disagreement follows model generation, not model family

Share of answered runs graded correct per question, by grader (majority of 10 grading rounds per answer). Nested rings = the graders agree on that question.



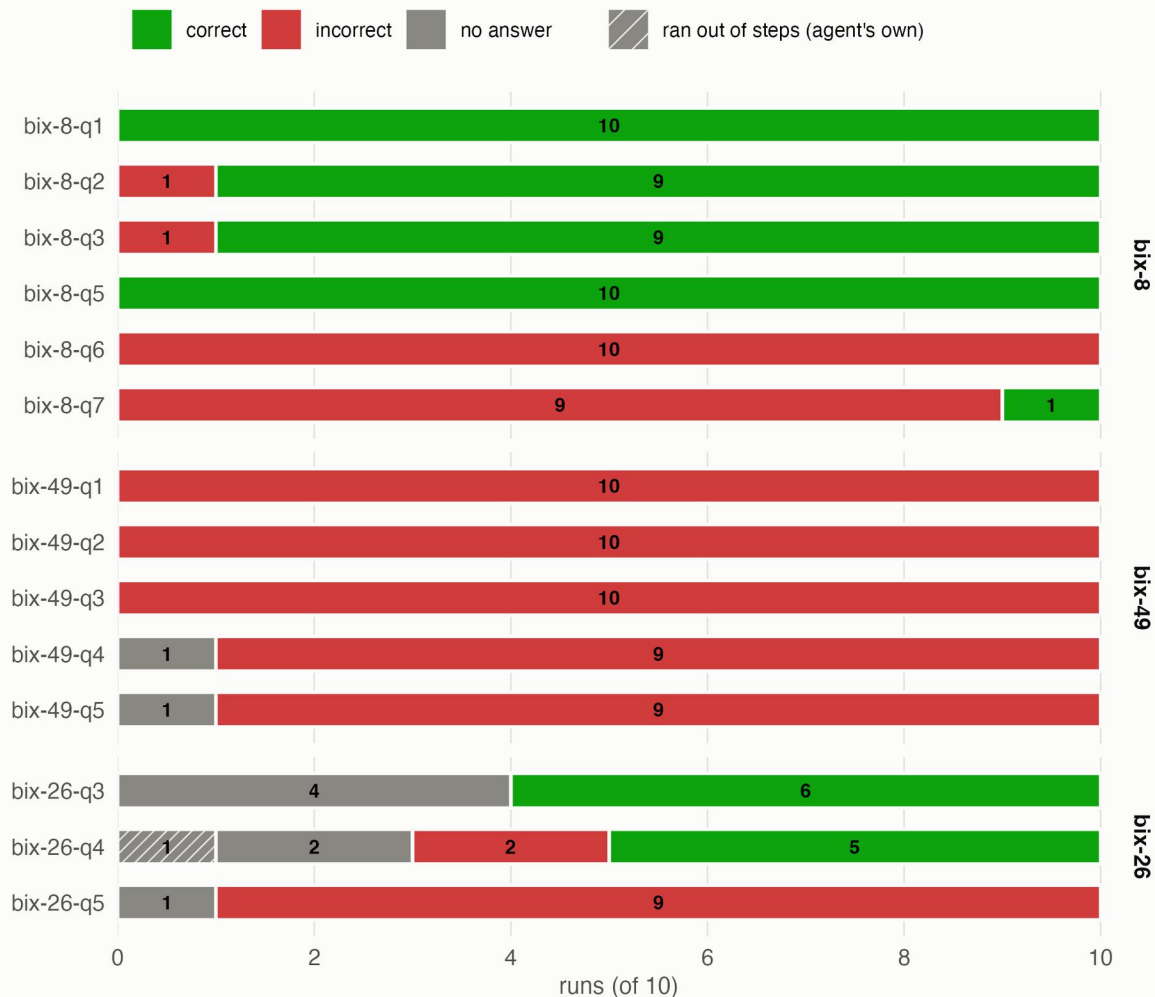
Same 130 answers under all three graders. gpt-5 is the primary grader for the remaining figures.

Two current-generation graders agree on all 130 answers. The shipped previous-generation grader contradicts itself on 14.6% of them.

FIGURE 2

No answer is its own outcome, not a wrong answer

10 replicate runs per question (claude-sonnet-4-5, temperature 1.0); correctness by gpt-5 majority vote.



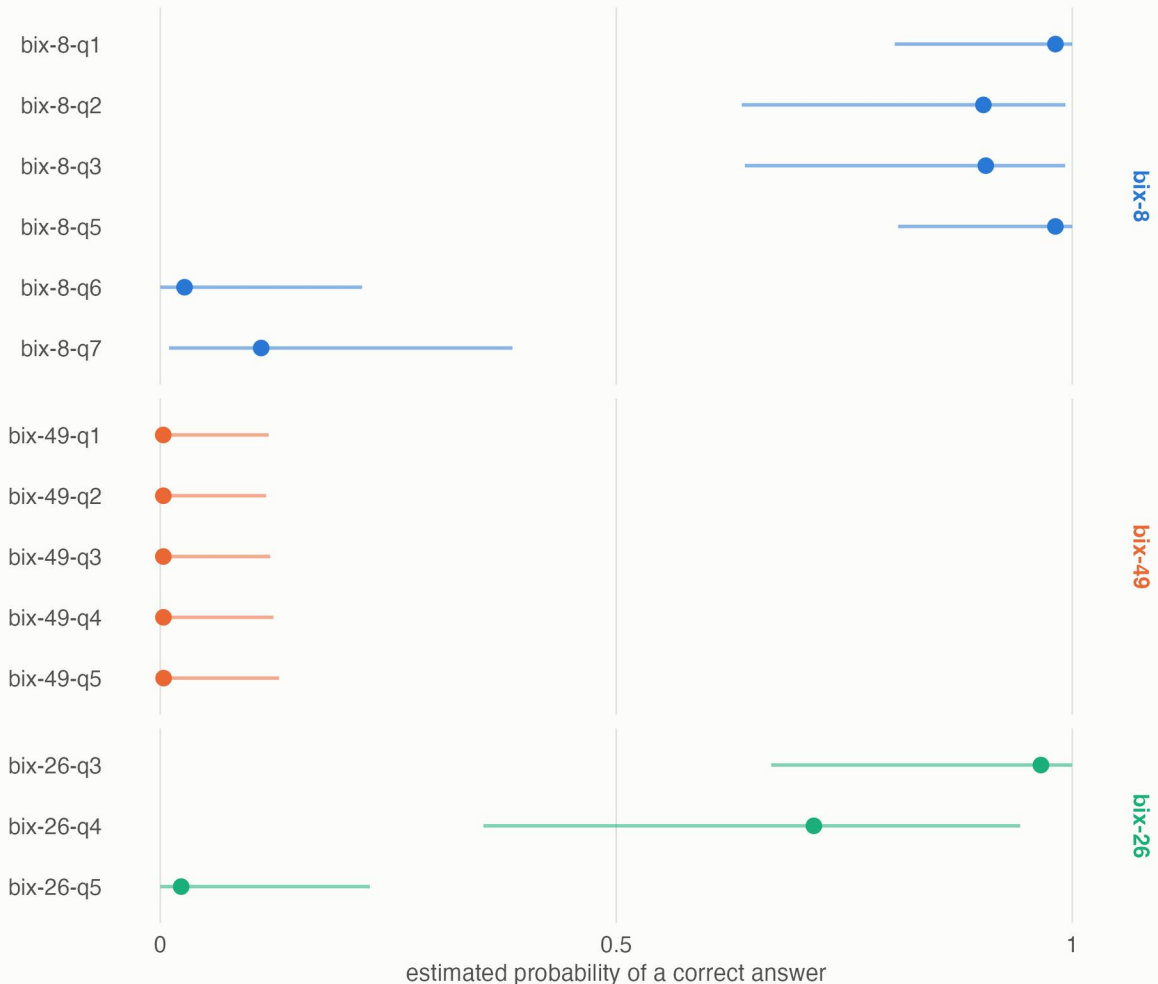
BixBench: 3 capsules covering 14 questions, 10 replicate runs each = 140 runs.

10 of 140 runs never answered - and 9 of those died following the harness's own package-install instruction.

FIGURE 3

Given an answer, correctness is a property of the question, not luck

Answered runs only. Dot = a question's estimated probability of a correct answer (posterior median); band = 95% uncertainty about that probability.



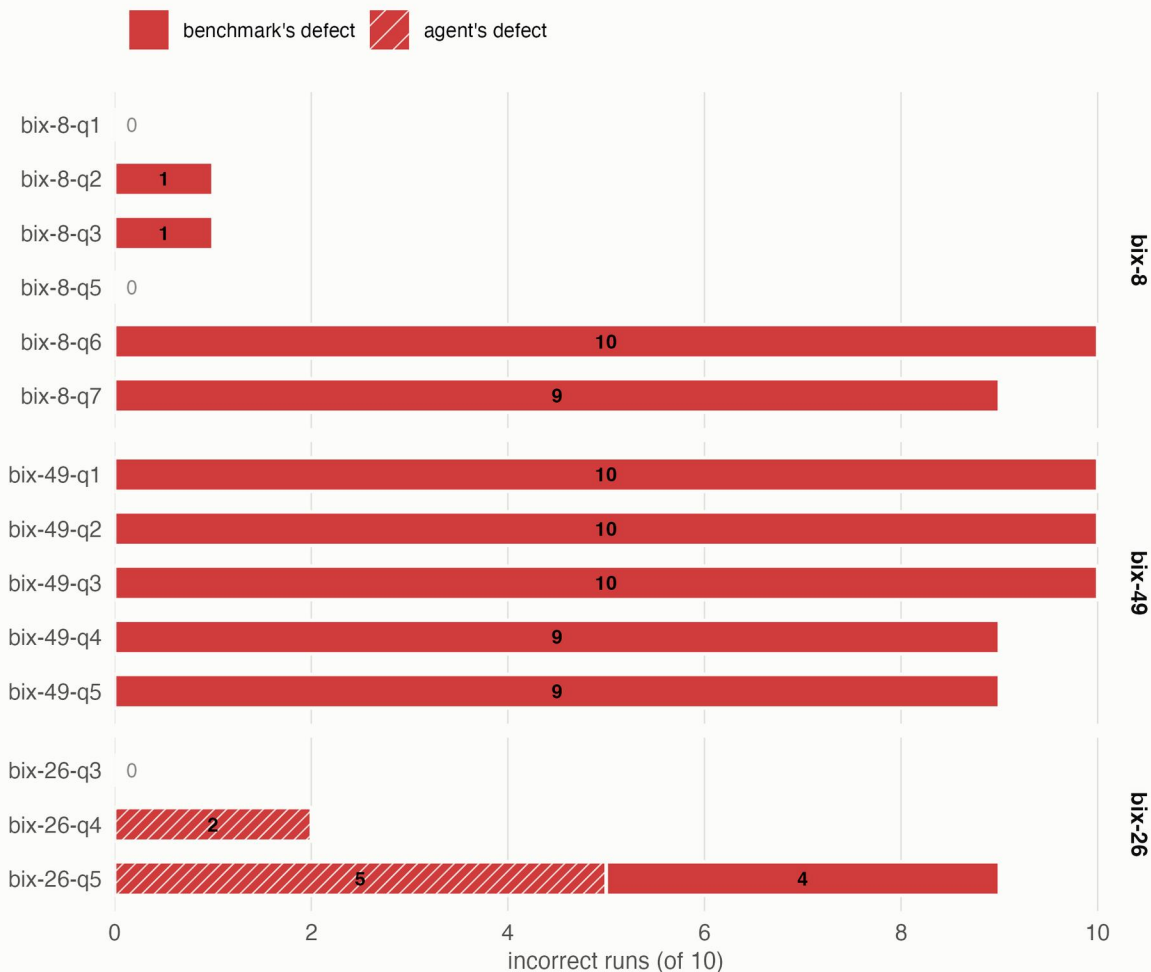
Bernoulli GLMMs, question nested in capsule; gpt-5 majority-vote grading. Question-level SD (stage 2): 3.2 log-odds.

Per-question success is near-certain or near-impossible, with almost nothing between: the outcome belongs to the question.

FIGURE 4

Most incorrect answers trace to the benchmark

Every incorrect run attributed by re-deriving both the answer key and the agent's answer from the raw capsule data (env/REVIEW_REPORT.md).



gpt-5 majority-vote grading. bix-8-q5's print-floor key is absent only because gpt-5 grades $p = 8.1e-194$ as satisfying " $p < 2.2e-16$ ".

Re-deriving every key from the raw data: most incorrect answers trace to the answer key or the question's wording, each cause verified.

READ THE FULL AUDIT

Every claim is **recomputable**

The writeup, the code, the grading data, and the case-by-case failure attribution are public - every recomputation is a committed script.

Writeup + code

github.com/Anqi-Dai/BixBench_eval

Interactive run audit

anqi-dai.github.io/BixBench_eval/results/figures/fig4_interactive.html

Data DOI

[10.5281/zenodo.22151974](https://doi.org/10.5281/zenodo.22151974)